

BenchDial

Voice agents that call, compare, and haggle for laboratory equipment repair
— so labs never overpay on a phone-priced job again.

01

Estimator

Voice + document → one confirmed
ScopePrint

02

Caller

Three negotiation styles, identical
job every time

03

Closer

Verified leverage, red-flag rule,
ranked award

Solo builder: Saadia Asghar · **Demo:** hacknation-lemon.vercel.app · **Repo:** github.com/Saadia-Asghar/Voice-Agent

PROBLEM

What we solved

Phone-priced markets hide the real cost until someone has the stamina to call five to eight operators. The Negotiator brief documents this for movers — identical work, 5.6× quote spreads, sight-unseen estimates that blow up later.

The same failures hit **laboratory equipment repair**:

Opaque pricing

Incomparable quotes

Call-out, calibration, warranty, and response assumptions differ wildly.

Unreliable estimates

Sight-unseen risk

Incomplete intake is why final bills jump after commitment.

No way to shop

One call at a time

Nobody has hours to run three honest negotiations while a centrifuge is down.

- Labs pay twice: experiment downtime *and* a misleading lowball.
- Software historically cannot dial the market — until voice agents.

AUDIENCE

Who it is for

Primary

Lab operations lead

Persona: Saadia Asghar, City Labs. Owns shared instruments. Needs the bench online without approving a quote that omits calibration or hides fees.

Secondary

Procurement / research ops

Need auditable vendor conversations and comparable terms — not a dashboard of apples-to-oranges numbers.

Vertical

Laboratory equipment repair (B2B) — same Negotiator loop that generalizes to movers, contractors, auto repair, and medical bills.

SOLUTION & FLOW

What the app does



USER ACTION	WHAT HAPPENS	WHY IT MATTERS
Upload report + lock scope	Canonical ScopePrint hashed and reused	Attacks sight-unseen blowups
Start live / fixture lanes	Three distinct negotiation styles	Mandatory challenge coverage
Review Deal Room	Itemized compare + red-flag + ledger	Price moves from leverage, not a script label
Open Award Memo	Ranked deal + citations	Judges see evidence, not a score alone

DIFFERENTIATION

Why this is a negotiator — not a form

Honesty Firewall

No invented bids

Server-side check_leverage must verify a competing term before the agent may speak it.

Provenance labels

LIVE vs FIXTURE

Fixtures never become RECORDED LIVE RUN. Only a signed webhook with a transcript can.

ScopePrint

Identical job every call

Voice and documents converge before dialing — configuration for the vertical, not a rewrite.

Human boundary

Recommend, don't bind

Award memo requires review. BenchDial cannot purchase or accept a contract.

STACK & ASK

How it is built

- **UI:** React + TypeScript + Vite — Estimator → Caller → Closer → Award
- **Voice:** ElevenLabs Agents (intake + Buyer/Closer) over WebRTC
- **Backend:** Supabase Auth, Edge Functions (tokens, HMAC webhooks, call-tools), Postgres evidence ledger
- **Logic:** Zod contracts, deterministic ranking, 30% below-median red-flag rule outside the LLM

One-line pitch

Turn one verified lab-repair brief into comparable AI voice negotiations and an evidence-backed recovery decision.

Links

Live:

<https://hacknation-lemon.vercel.app>

GitHub:

<https://github.com/Saadia-Asghar/Voice-Agent>

Team:

HN-6590 · Solo · Saadia Asghar

BenchDial recommends. A human decides. The bench gets back online.